F1 Hankel Matrices: Hand and Julia

Numerical Validation Progress Record | 27 October 2026

Checkpoint: Numerical agreement verified for the selected scalar example. **Open:** Saved approximate assertions and an unused wrapper defect. This record does not assert completion of the full F1 series or Experiment 001.

1. Scope and continuity

This session extended the hand-built F1 Hankel example into a Julia/Pluto construction. The hand reference was already documented in the *F6 Prediction Target + F1 Hand Hankel Progress Record* of 27 October 2026, section 3. Earlier prerequisite and initial-state work were not repeated. The current result concerns one noiseless scalar dataset.

Checkpoint	Status	Evidence / boundary
Hand sequences and matrices	**Verified**	Existing independent reference, dimensions, row order, and time windows.
Julia Hankel construction	**Verified**	Both blocks and the stack agree within the existing tolerances.
Floating-point interpretation	**Recorded**	The user connected nearly identical results and a false exact comparison to the need for `isapprox()`.
Saved checks and helper cleanup	**Open**	Explicit assertions and the recursive `Behavior` indexing method remain follow-ups.

2. Dataset and construction

The discrete-time, noiseless scalar example is

$$x_{k+1} = 0.8x_k + 0.5u_k, \qquad y_k = x_k, \qquad x_0 = 1.$$

No physical units or physical sampling interval are assigned. There are $N = T = 6$ recorded input/output pairs and Hankel depth $L = 3$:

$$u_{0:5} = [\,0,\ 1,\ -1,\ 1,\ 0,\ 1\,],$$
$$y_{0:5} = [\,1,\ 0.8,\ 1.14,\ 0.412,\ 0.8296,\ 0.66368\,].$$

The constructor advances each length-L window by one sample:

$$H_3(u), H_3(y) \in \mathbb{R}^{3\times 4}, \qquad H = \begin{bmatrix} H_3(u) \\ H_3(y) \end{bmatrix} \in \mathbb{R}^{6\times 4}.$$

Columns cover times 0:2, 1:3, 2:4, 3:5; input rows precede output rows. Stored states include time 6, while recorded outputs stop at time 5. The Hankel blocks use the aligned input/output records.

3. Numerical agreement and learning reflection

Comparison	Exact equality	Approximate comparison
Input block $H_3(u)$	True	Exact match to the hand reference.
Output block $H_3(y)$	False	Passed with rtol = atol = 10^{-12}.
Combined matrix H	False	Passed with the same tolerances.

$$\max_{i,j} \left| H_{ij} - H_{ij}^{\text{hand}} \right| = 2.220446049250313 \times 10^{-16}.$$

Learning reflection. The results were nearly identical. Small floating-point discrepancies made the exact Boolean comparison false. This highlighted why it is important to use `isapprox()` with explicit tolerances. Reference values were not rounded and tolerances were not changed to obtain a pass.

4. Actual validation and its limits

The following are outcomes of the read-only Julia review on 27 October 2026. They were not rerun for this documentation update.

Check	Outcome	What it establishes
Saved notebook cells	18 completed	Non-setup cells in the reviewed snapshot executed. Setup and an empty cell were skipped; later additions were display-only or empty.
Independent Hankel checks	**10/10 passed**	Hand references, block/stack shapes, matched sequences, depths $L = 1$ and $L = N$, integer data, and view inputs.
Direct package suite	**15/15 passed**	One smoke assertion, five original simulator assertions, and nine simulator regression assertions. This suite does not contain the notebook-local Hankel tests.
Isolated `Behavior` indexing	**Failed**	Indexing raised `StackOverflowError`; the method calls itself. The wrapper is unused by the current Hankel calculation.

The direct package command was:

```
julia --project=.  --startup-file=no --compiled-modules=no test/runtests.jl
```

Diagnostic observations. The saved `H == hand_H` evaluates to false, but the following `hand_H` expression hides the Boolean in the cell output. Depth probes returned an empty 0×7 matrix at $L = 0$, an empty 7×0 matrix at $L = 7$, and `ArgumentError` at $L = 8$. The accepted depth domain should be documented before broader reuse.

Execution boundary. Direct checks emitted Markdown/Test import warnings. The notebook setup/instantiation cell and full Pluto UI were not executed. The earlier extraction-stage CI command `Pkg.test(; allow_reresolve=false)` passed 15/15; that historical result is separate from this review. Neither this example nor its tests establish general trajectory spanning or persistent excitation.

5. Follow-up work

1. Save explicit approximate assertions for both blocks and the combined matrix, plus their expected shapes, using the existing tolerances.
2. Fix `Behavior` indexing by delegating to the stored data, or defer the unused wrapper. Its failure must not be hidden by the successful Hankel checks.
3. Complete the notebook's comparison-outcome text. The user's floating-point interpretation is now recorded here and in the two repository progress records.
4. Before reusing or extracting the constructor, define its input/depth contract and preserve the independent hand-reference checks as regression tests.

6. Evidence and reporting

Item	Record
Repository	`darienperez/DataDrivenControl`; working branch `foundation/f1-hankel-matrices`, based on committed checkpoint `b8930e1`. The current work remains uncommitted.
Implementation	`notebooks/F1_hankel_matrices.jl`; shared simulator in `src/simulate_lti.jl`.
Continuity	`experiments/F1_hankel_matrices.md` and `experiments/F1_benchmark.md`; hand source: `F6_F1_Progress_Record_2026-10-27.pdf`, section 3, page 2.
Time and external logs	No focused-time total was supplied for this record. No calendar, tracker, or Google Task changes were made in this update.

Record status: The selected hand-versus-code numerical checkpoint is verified. Saved validation and wrapper cleanup remain open. Broader F1, Experiment 001, and linked-task completion are not asserted.